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Review

Agriculture as Energy Prosumer: Review of Problems, Challenges, and Opportunities

Piotr Sulewski and **Adam Wąs** *

Institute of Economics and Finance, Warsaw University of Life Sciences—SGGW, Nowoursynowska 166, 02-787 Warsaw, Poland; piotr_sulewski@sggw.edu.pl
* Correspondence: adam_was@sggw.edu.pl

Abstract: The issue of energy in agriculture is complex and multifaceted. Historically, agriculture was the first producer of energy through the conversion of solar energy into biomass. However, industrial development has made agriculture an important consumer of fossil energy. Although the share of agriculture in the consumption of direct energy carriers is relatively small, today's agricultural producers use many inputs, the production of which also consumes much energy, mainly from fossil fuels (e.g., synthetic fertilizers).The food security of the world's growing population does not allow for a radical reduction in direct and indirect energy inputs in agriculturer. Undoubtedly, some opportunities lie in improving energy efficiency in agricultural production, as any waste of inputs is also a waste of energy. In addition to improving efficiency, the agricultural sector has significant opportunities to consume energy for its own use and for other sectors of the economy. Biomass has a wide range of applications and plays a special role here. Other forms of renewable energy, such as increasingly popular agrovoltaics, are also important options. When analyzing the place of agriculture in the energy system, it is therefore worth seeing this sector as a specific energy prosumer, which is essential in the energy transition process. Such a point of view is adopted in this study, which attempts to identify the determinants of agriculture as a consumer and producer of renewable energy.

Keywords: energy efficiency; fossil fuels; bioenergy; agrovoltaics; energy prosumer

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1. Introduction

The results of archaeological research conducted in various parts of the world indicate that the history of agriculture spans at least 12,000 years since various nomadic tribes began a sedentary lifestyle and mastered the ability to cultivate several basic plant species and raise animals [1]. The transition from gathering and hunting to agriculture as the primary means of obtaining food was crucial to the emergence of the first civilizations, and agriculture became the main source of energy and materials for preindustrial societies [2,3]. Agriculture can thus be seen as the first energy generation sector ever developed by man [3]. Sources of energy used in agricultural production processes in the preindustrial era were mainly organic inputs based on solar energy (biomass) and animal pulling power (fed with biomass) [2,4]. As part of the photosynthetic processes, the energy of the sun's radiation was converted in a targeted manner into energy contained in food, feed, and fibers (i.e., biomass) [5]. The Industrial Revolution changed this situation; fossil fuels and minerals became the main source of energy and materials for people [2,6,7]. Also, in agriculture itself, with industrial development, inorganic inputs (fertilizers, machinery) and energy based on fossil fuels have taken over the dominant role in food production processes [2,4], and agriculture has gone from being an energy producer to a significant energy consumer, dependent on fossil resources [3,5,8].

More than 200 years of industrial development have made a large part of the world's population richer than ever before. At the same time, industrialization has contributed to the overexploitation of natural resources and the accumulation of greenhouse gases, which consequently began to raise questions about the viability of humanity [9,10]. For many years, the negative consequences of economic development based on non-renewable resources were barely recognized. It was only in the 1960s and 1970s that discussion began about the fact that the existing economic growth model based on intensive exploitation of natural resources (including energy resources in particular) is a source of environmental hazards and internationally growing social inequality [11–15]. Undoubtedly, one of the factors that contributed to the change in thinking about non-renewable raw materials was the energy crisis of the 1970s, which, although it was politically rather than environmentally motivated, nevertheless made authorities and societies realize that access to energy resources is not unlimited [16,17].

A significant contribution to the processes of environmental degradation and consumption of natural resources is also made by agriculture, which, in the process of intensification of production, has significantly increased its productivity and provided food for the growing population of the globe [18,19], but, at the same time, has contributed to the generation of many adverse environmental effects [20]. These effects are manifested in, among others [21]: eutrophication (mainly the impact of excess use of nitrogen fertilizers (globally estimated excess is 37%) and phosphorus (estimated excess is 8%), loss of biodiversity (among others, as a result of the use of pesticides, the global consumption of which reaches 335,000 tons per year), soil degradation (e.g., as a result of erosion, mechanical treatments), excessive water consumption (40% of total water consumption), greenhouse gas emissions (globally about a 13% share of total emissions, but excluding energy carriers, which in IPCC statistics [22] are attributed to the energy sector), or air pollution from dust and odors. Dependence on fossil fuels means that globally, GHG emissions from energy consumed in agriculture reached (in 2018) 0.9 billion tons of CO_2 eq. (against total emissions from agriculture of 9.3 billion tons in 2018) [23]. Between 1990 and 2018, on-farm GHG emissions increased by one-quarter [24]. Considering the entire food system, it can be pointed out that about one-third of the total global GHG emissions is related to the agri-food sector [25]. Annual emissions related to energy consumption in agriculture are estimated to have increased by 7% between 1990 and 2019, with a marked decline in developing countries and a slightly larger increase in developing countries [26].

Agricultural pollution processes can be seen as a result of the sector's low energy efficiency, as each of the phenomena above is associated with unproductive energy use [3,5,8]. It should be borne in mind that energy consumption is not limited to direct consumption of energy carriers (liquid and solid fuels, electricity, heat) but also includes indirect consumption, consisting of energy embodied in different inputs of production, among which a special place is occupied by synthetic fertilizers, accounting for up to half of the total energy consumption in agriculture [5,27,28].

Moreover, it is estimated that up to 30–40% of the food produced globally is wasted [29–31] (at various stages of the supply chain, including about 13–15% before store shelves [31,32]). This means that the energy previously used to produce this food is also wasted [33,34], and these losses magnify the scale of the negative impact of agriculture (and the entire food system) on the environment [35,36]. So, when considering food system management, it is essential to remember that "food is energy" and "food lost and wasted is energy wasted" [36]. It is worth mentioning that according to available estimates, GHG emissions associated with farm-stage food waste reach 4% of all anthropogenic greenhouse gas emissions and 16% of agricultural emissions [31].

There is no doubt, therefore, that agriculture is an important consumer of energy (even if it is not visible in statistics), and inefficient use of energy (direct and indirect) generates many negative effects (such as eutrophication resulting from inefficient use of nitrogen and phosphate fertilizers). On the other hand, however, agriculture is a sector that is particularly predisposed to the generation of various types of renewable energy, particularly biomass-based, which means that the sector can be seen as a kind of "collective prosumer" of energy, able to generate energy for its own needs as well as those of other sectors [5,37,38]. Data in the REN21 [39] shows that currently, on a global scale, only about 15.4% of the energy consumed by agriculture comes from renewable sources, including only about 5.3% from modern bioenergy (excluding wood). The potential of agriculture in energy production is being exploited to a small extent today. The challenges of the energy transition are prompting both a search for ways to improve the efficiency of energy use and ways to replace fossil fuels with renewable sources. In pursuit of the UN's global Sustainable Development Goals, including "affordable clean energy" [32], it seems necessary to change the role of agriculture in the modern economic and energy system and move from the position of an energy-inefficient consumer that generates many externalities to the role of a prosumer making efficient use of the energy it produces and assisting other sectors of the economy (in the energy supply and realizing environmental goals). In this context, the article aimed to identify and assess the prospects, possibilities, and chances of agriculture as a "collective energy prosumer". It consists of three sub-parts that characterize the logic of prosumerism, i.e., a discussion of the agricultural sector as an energy consumer, the role of agriculture as an energy producer, and a discussion of issues related to energy efficiency in this sector. These considerations were carried out in the context of the basic assumptions of the prosumer energy role in an energetic system.

2. The Prosumer in the Energy System

The traditional model of the economy assumes the existence of two basic participants in economic exchange, i.e., producers providing goods and services and consumers purchasing these goods. Alongside this division is the concept of the "prosumer", where the boundaries between producer and consumer are blurred, as this category of people produces goods and services for their own or community members use [40]. The concept of prosumers has recently gained popularity in the energy industry and digital economy [40]. However, this concept was developed much earlier by A. Toffler [41], who argued that consumers are a phenomenon of the "industrial age". While societies move towards the "Post-industrial Age", the number of "pure consumers" is decreasing, and their place is increasingly taken by "prosumers", who produce goods and services for their own use [42]. In practice, prosumers exemplify the redefinition of traditional economic roles to produce value for themselves or others [40]. The concept of prosumerism refers to such economic concepts as "sharing economy", "collaborative consumption", "co-creation", "co-production", "self-production", or "consumerchant", and in many respects is still underdetermined and sometimes lacks a clear distinction between different categories of market participants and prosumers [43]. In principle, however, the logic of prosumerism also fits into the sustainable development paradigm [44,45] and the "circular economy" [46]. Examples of the spread of prosumerism can be pointed out, among others, in such industries as healthcare, education, fashion, art, 3D printing, Bitcoin mining, or agriculture, but the most significant impact of the prosumerism phenomenon is on the energy industry [40,44,47–49]. Among the examples of prosumerism in rural areas are food prosumers [50]. Generally, it can be assumed that energy prosumers are individuals who produce and consume energy simultaneously. Prosumerism represents a historical shift from large-scale companies to dispersed energy generated by active consumers [51]. The category of energy prosumers, like the category of prosumers in general, is heterogeneous and can be subdivided [52]: "individual investments" (the most common way that homeowners become a prosumer is installing a PV plant); "collective prosumers in one building"; "community investments" (the most diverse group of prosumers); "participation in company-led projects" (individuals' participation in regional renewable energy projects is based mainly on financial participation). Another classification indicates such categories of prosumers as households, public entities, small enterprises, and the tertiary sector [53]. An important characteristic of prosumers is to be involved in the energy community and represent its citizens; thus, for example, industrial companies producing parts for their own needs are not considered prosumers because they do not directly represent citizens [52].

An essential aspect of the existence of energy prosumers is that their activity should promote the goals of sustainable development [44,45,47] and thus, in practice, be based on renewable energy sources and be consistent with the UN's Sustainable Development Goal "affordable clean energy" [32]. Typically, the idea of prosumerism in the energy field is clearly associated with the consumption and production of renewable energy [54]. In practice, it is difficult to imagine a farmer being able to generate energy from fossil fuels, so the article assumes that being a prosumer in the case of farmers is related to the generation of renewable energy, which seems to be consistent with the point of view prevailing in the literature [5,39,47,55].

Increasing sustainability and reducing dependence on fossil fuels has a natural impact on the entire economic system and society; however, success in implementing green and circular economy principles depends on various factors [47,56–58]. However, this transformation will not happen without increasing the use of RES energy, for whose generation in a distributed system prosumers are beneficial [44,45,59]. According to Ritchie et al. [60], on average worldwide, the share of renewable resources in primary energy consumption (transport, electricity, heat) was 14.56% (in the EU, it was almost 22%; in OECD countries, 16.5%; in non-OECD countries, 13.4%). On average worldwide, about 47% of this energy comes from hydropower and almost 26% from wind, just over 18% is solar energy, and only 8.7% is from other renewable resources, which include bioenergy arising from biomass (excluding traditional biomass) and also geothermal, wave, and tidal energy [60].

Among the main factors that stimulated the development of energy prosumption in the EU were the falling costs of renewable energy sources and the support of this form of activity by many governments (in the form of subsidies or net-metering systems), as well as the growing environmental awareness of citizens [52,61,62]. Nonetheless, it is currently difficult to determine the exact number of energy prosumers, even in EU countries that have particularly intensively supported the development of renewable energy sources through prosumerism in recent years. The EEA study [52] shows that between 2015 and 2020, for example, the number of PV prosumers in the Netherlands increased from less than 500,000 to more than 1 million, in Portugal from 3,000 to more than 30,000, and in Poland from 51,000 in 2018 to 847,000 in 2021. In general, however, it can be said that the development of prosumerism in energy generation to date has been based mainly on photovoltaic panels generating electricity installed on rooftops and on the ground, and to a much lesser extent on wind power [52,53]. Some estimates suggest that prosumers in EU countries relying on these two sources can meet 30 to 70 percent of their electricity needs (depending on the country) and their total heat needs relying on electricity, biomass, and solar heat [52,53].

It is worth noting that in the context of prosumer energy development opportunities in the area of agriculture, in addition to initiatives involving the commissioning of energy sources on individual farms, there are also opportunities for collective actions such as "hybrid energy-agriculture cooperative" (renewable energy communities and cooperatives), which can improve the use of renewable energy produced by farmers [63]. Such cooperatives should have several different renewable energy sources, interconnectedness of members, bi-directionality of energy flows, and greater flexibility than facilities operating individually. Developing such cooperatives can help unlock bioenergy's untapped potential in local communities and support the decarbonization of agriculture and other sectors. It is worth noting at this point that one of the factors in the development of renewable energy cooperatives is social capital [64]. Although collective energy presumption initiatives have become a fairly widespread form of energy generation in rural areas, their development still faces numerous barriers (political, legal, economic, and social-cultural conditions) [65].

The development of prosumer energy is mainly based on the following technologies [53]:

- Rooftop or ground-based solar PV (for electricity generation): photovoltaic (PV) technology represents prosumers' most pervasive energy generation technology. Its principal advantage is ease of installation, while its primary disadvantage is dependence on weather conditions. Installation of PV on the ground necessitates the occupation of additional space, rendering it a relatively uncommon solution in households. However, it is more readily implementable in rural areas.

- Wind turbines (to generate electricity): These are based on wind energy and can be mounted on the roof or on the ground. Ground-mounted wind turbines (like ground-mounted PV panels) are used in pro-supply power generation and commercial solutions (installed for the sole purpose of selling energy).
- Solar thermal (to generate thermal energy): Solar thermal energy can be used to capture the heat from the sun and is usually used in conjunction with other sources of renewable energy (such as PV).
- CHP (to generate electricity and heat): CHP can be seen as a prosumer technology if the prosumer himself produces the energy carrier used. In practice, this applies only to biomass energy, especially in rural areas where sufficient biomass can be obtained. The use of this technology is unlikely in the case of individual households but is often used on farms with a biogas plant.
- Biomass boiler (to generate thermal energy): Similarly to CHP, this technology can be seen as prosumer if the biomass used is produced by the prosumer, which in practice limits its use to rural areas.
- Heat pumps (to generate heat energy): this technology is based on extracting energy from the environment (air, ground, or water) and can be used in both urban and rural areas.

It is worth adding that the effective operation of any system based on renewable energy sources (both in agriculture and outside) is strongly dependent on the conditions of the region, which consist of, among others, economic characteristics (conducted activities), road infrastructure, energy infrastructure (transmission networks), population density, etc. [38].

The literature offers examples of farmers functioning as consumers and producers of energy at the same time [66,67]. Pereira et al. [67] point out that nowadays, it is not enough to use renewable energy sources alone, but also appropriate energy management is important, for which various types of solutions based on smart farming are useful. The analysis presented here shows that integrating modern management with one's own energy source, in the case of a farm from Portugal discussed by the authors, made it possible to reduce energy consumption from the grid by more than 83%. Of course, solutions based on photovoltaics (agrovoltaics) are dependent on insolation, so in different parts of the world, the effects will vary. Agricultural biogas, whose production does not depend on the weather but only on the supply of biomass, is pointed out as a stable source of energy for farmer-prosumers. Biogas production does not always have to mean a huge investment project; in rural areas, especially in developing countries, small biogas digesters are often installed to provide gas for the family's needs; the cost of such a solution does not exceed the cost of acquiring natural gas [68,69] In developed countries, where agriculture is expected to produce biogas not only for its own consumption but also for other sectors of the economy, biogas projects involve large investments. For example, a country where agricultural biogas production has been particularly intensively developed is Germany, where the development of this sector, however, was intensively supported by public funds and also relied heavily on energy crops, which has become the subject of controversy [70]. Nevertheless, 9500 biogas plants supply about 50 TWH of final energy to the German energy system [71]. In contrast, a slightly different model was developed in Denmark, where large biogas plants were built mainly using waste biomass from many farms and the food industry [72]. The Danish model puts manure utilization first and energy production second, but this leads to synergistic effects and provides multidimensional benefits for farmers, the environment, and energy consumers [72,73]. In general, however, high investment costs will make the success of a biogas venture depend on the level of support in practice. Unfortunately, biomass power generation costs are decreasing at the slowest rate of all RES. Between 2010 and 2023, the LCOE (levelized cost of electricity) decreased for bioenergy by 14%, compared to 90% for solar PV, 70% for onshore wind, 63%

for offshore wind, 31% for geothermal, and 33% for hydropower [74]. A review of some examples of anaerobic digestion of agricultural waste for biogas generation can be found in [75]. As for examples of farmer use of wind power, this is a fairly popular technology in some parts of the US, where small-scale wind turbines have been built since the 1930s. It is increasingly common for third-party developers to erect turbines on farmland, but this is a far cry from the prosumer model. In recent years, such a model has been developing in Germany, for example [76]. Although many successful examples might be presented, problems are also pointed out. One is the form and level of financial support provided to prosumers. In Poland, for example, until 2022, prosumers could take advantage of net metering, which contributed to the rapid growth of solar PV masts. The change of the system to net-billing has significantly changed the conditions, and according to Lakomiak's analysis [77]; this limits the use of the potential of renewable energy production on orchard farms (both by individual prosumers and by energy cooperatives).

3. Agriculture and Energy

Although awareness of the negative impact of agriculture on the environment in societies began to build relatively recently [78–80], the issue of energy in agriculture as such was probably first scientifically considered as early as the 19th century [81]. However, energy issues in agriculture generally remained outside the realm of scientific analysis until the 1970s, when the energy crisis prompted consideration of the energy efficiency of various sectors of the economy [82]. As Martinez-Alier [3] points out, one of the first studies to accentuate the problem of energy efficiency in agriculture was that of Pimentel et al. [82], which showed that "*energy efficiency of corn production in Iowa or Illinois was lower than that for the traditional milpa corn production system of rural Mexico*" [3]. The starting point for this type of consideration was H.T Odum's book [83] *Environment, Power and Society*, in which the author pointed to modern agriculture as "farming with petroleum" [3]. Nowadays, it is accepted that agriculture has transformed from an energy producer to an energy consumer [3,5,8]. According to this vision, agricultural activity, by intervening in the ecosystem, provides a "*surplus of photosynthetic energy geared directly to human nutrition*", but generally, "*energy made available by agriculture is less than the amount that would be naturally available from the same territory*" [3]. Some calculations indicate that back in the early 1940s, the average U.S. farm produced about 2.3 kcal of food energy from 1 kcal of fossil energy; in 1974 the ratio was 1:1, and by the beginning of this century, it was already 1:3, meaning that the production of 1 kcal of food energy involved the consumption of 3 kcal of fossil energy (and this value does not include the distribution of transport, etc.) [84]. Including operations throughout the supply chain, this number rises to as much as 10–15 kcal used to produce 1 kcal of food energy [84,85]. These values indicate the low energy efficiency of the energy transformation system in agriculture, which generates the need to look for solutions that would lead to a change in this situation. It is worth noting that the complex nature of the links between production and energy consumption in agriculture makes different approaches possible in assessing the energy balance of the sector [85]. The energy issue in agriculture is undoubtedly complex, and the energy sector and agriculture are strongly interlinked. Some authors even use the term "energy–agriculture nexus" [5,86]. This means that energy is needed to power various agricultural activities at different stages of production. On the other hand, agriculture can generate energy (biofuels, biogas) by processing the products and by-products or wastes generated in farming activities, which links the energy sector to agriculture [5,86]. In addition to converting biomass into more valuable energy, agriculture can directly generate energy from the sun and wind.

The existing interrelationships between the different aspects of the energy cycle in the agricultural sector are shown in Figure 1. This Figure also reflects the issues covered in the study.

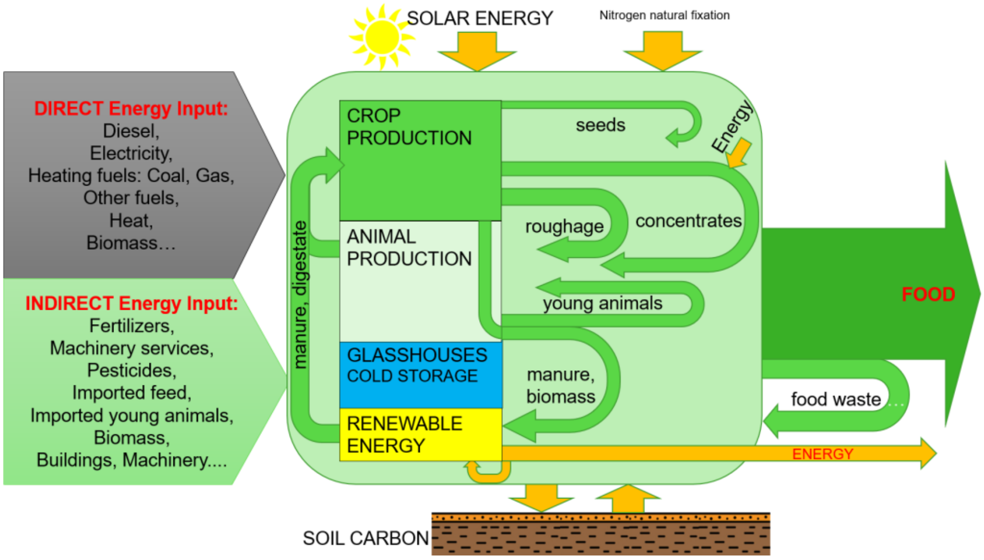

Figure 1. Energy flows in the agricultural sector. Source: own elaboration.

4. Agriculture as an Energy Consumer

4.1. Direct and Indirect Inputs of Energy

As indicated earlier, economic development and industrialization have transformed agriculture from an energy producer to an energy consumer [3], and fossil fuel energy has become one of agriculture's primary inputs [55,87]. Energy consumption in agriculture varies considerably depending on the type of crop/activity, geographical location, climate, production practices, agricultural machinery, or production intensity [88]. It also varies according to the economic situation—in general, it can be observed that the better the economic situation, the higher the energy consumption in agriculture [89].

Energy in agriculture is consumed both directly in the production processes of agricultural commodities (e.g., fuel for machinery) and indirectly for the production of other inputs needed in agriculture (e.g., fertilizers) as well as in food processing and distribution processes [5].

Direct energy inputs, including the energy of fossils, used in the agricultural process represents the sum of the consumed electricity and solid, liquid, and gaseous fuels [27]. Direct energy refers to the energy used on-site (e.g., diesel) and usually reflects the heating value of the energy carrier and omits energy connected with the production and distribution of the carrier. In the case of diesel, a liter of fuel carries about 38 MJ of energy (gross caloric value). Still, additionally, its extraction, processing, and delivery to the farm require 9.1 MJ, which is usually skipped in the calculation of farm energy consumption [90]. In general, direct energy use is assumed to include energy inputs used directly in production processes up to the "farm gate" stage. It includes electricity consumption, refined petroleum products (mainly diesel), natural gas-based fuels, and wood chips [27,28]. The indirect cost (inputs) includes energy accumulated in inputs consumed in agricultural production processes (including energy carriers used for manufacturing of production means, including fertilizers, pesticides, farm machinery, and farm buildings [27,91]. It is estimated that more than half of the energy used in agriculture is related to indirect energy uses, mainly from nitrogen [91–93]. Accurate estimation of indirect energy use is more complicated than determining direct energy consumption, especially in the case of machinery and buildings, as this would require exact construction data under conditions of huge variation in technical solutions [91,94]. It is even challenging to categorize the energy consumption contained in farm-generated biomass. For example, straw resulting from agricultural production can be treated as heating fuel (energy carrier with a specific GCV of up to 18 MJ/kg) [88]. Still, it can also be consumed on the farm as cattle bedding or plowed in the field for soil improvement. It is worth mentioning that this straw is produced during the cultivation process as a result of direct (e.g., diesel for tractor) and indirect (e.g., fertilizer inputs) energy use and as a result of solar energy accumulated through photosynthesis [5]. Energy derived from biological and renewable sources (biomass) can be classified as bioenergy [95]. The biomass produced and subsequently used on the farm in various production processes represents a potential indirect energy input (bioenergy), which is usually overlooked in analyses of energy consumption in agriculture. This may refer to the straw mentioned above, organic fertilizers, roughage, grains used for concentrated feed, seeds, and young animals. Biomass used to produce heat or electricity should be classified as direct energy inputs. Synthetically, the basic categories of energy inputs in agriculture are summarized in Table 1.

Table 1. General classification of energy inputs in agriculture. Source: own elaboration based on: [27,85,91,93].

Categories of Energy Inputs in Agriculture		
Direct Energy Inputs	**Potentially Indirect Energy Inputs**	**Indirect Energy Input**
Diesel Electricity Heating fuels: Coal, Natural Gas, Heat Wood Biomass for energy production e.g., food waste, by-products of the food industry for biogas plant	Seeds Concentrate feed Roughage Young animals Organic fertilizers (e.g., manure, farm leftovers)	Fertilizers Pesticides Farm buildings and farm infrastructure development Machinery and other equipment Services, e.g., machinery contractors, seeds preparation, drying cereals Industry-based animal feed e.g., molasses, beet pulp

It is estimated that in the category of direct energy cost, the share of agriculture in final energy consumption is at about 3%, while the share of the entire agri-food sector is already at about 30% [55]. In the EU, it is also 3% on average, but there are significant differences between countries in this regard—for example, in the Netherlands, it is over 9%, and in Slovakia, it is 1.3% [96]. A slightly smaller average share of agriculture in global energy consumption (directly) is indicated by FAO data (Figure 2), which shows that the global share of agriculture and forestry in 1971–2009 did not exceed 3%, but the situation varied strongly, on average, between world regions. By far, the narrowest share throughout the observation period occurred on average in South America (variation in the range of 4–6%). On the other hand, the lowest was observed in developing countries and North America. It is also worth noting that the indicator in question grew quite steadily until the early 1970s, after which a slight downward trend began (mainly in Europe and South America). When analyzing the changes in the share of agriculture in energy consumption, it should be remembered that they are also the result of changes in energy consumption in other sectors of the economy.

Regarding changes over time, an interesting observation is provided by an analysis of U.S. data, which shows that energy consumption in U.S. agriculture grew dynamically until the late 1970s, after which it began to decline, but what is particularly interesting is that this was primarily influenced by a decline in direct energy use (Figure 3). It can be assumed that the trend was also similar in other developed countries.

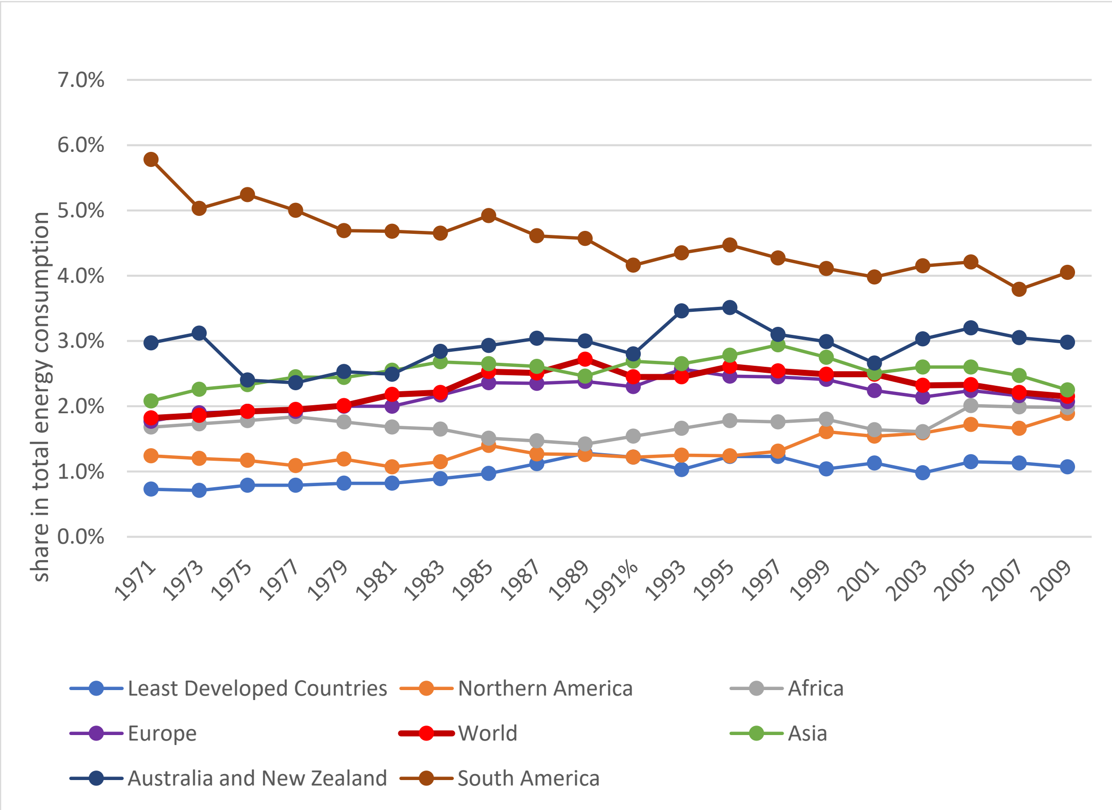

Figure 2. Share of total energy (direct) used in agriculture and forestry. Source: elaboration based on FAOSTAT processed by Our World in Data (license CC BY) [97].

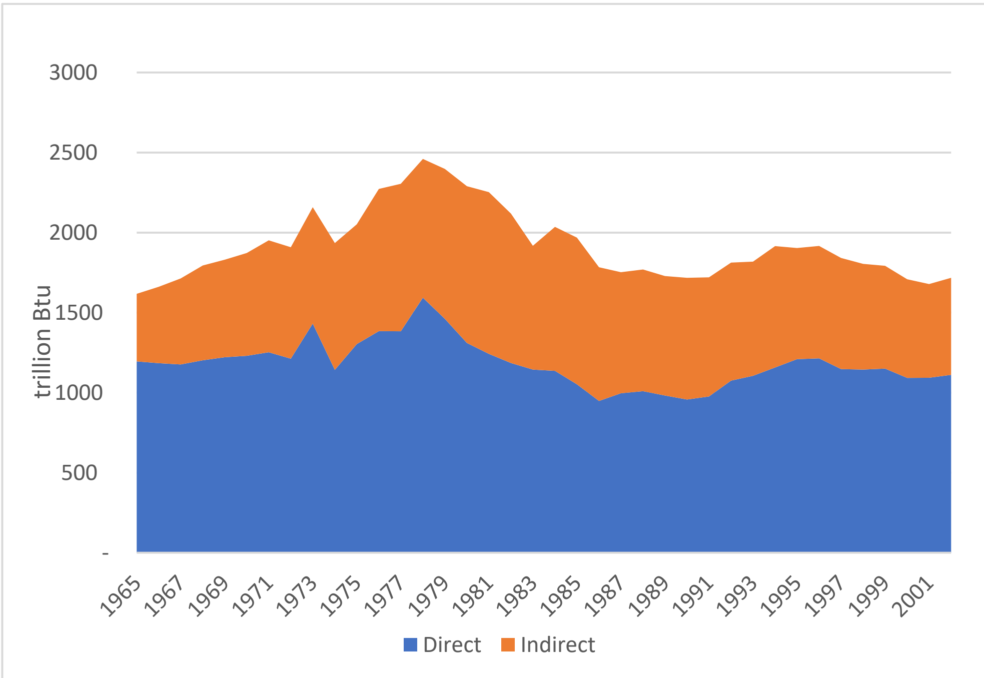

Figure 3. Historical trends in direct and indirect energy use in U.S. agriculture in the period 1965–2001. Source: elaboration based on [98].

Between 2011 and 2021, global direct energy consumption in agriculture increased by about 12% [39]. More than 85% of the energy consumed globally by agriculture came from non-renewable sources in 2021 [99]. However, it should be noted that the share of renewable energy in the energy consumed by agriculture between 2011 and 2021 increased from 10.8% to 15.4%. Thus, it is possible to observe a positive trend in changes in the share of RES. Still, given the scale of the challenges associated with the UN's Sustainable Development Goals (in particular, the "clean energy" goal) or the EU's drive for climate neutrality and the Green Deal assumptions [100], the observed level should be considered far from sufficient. Climate neutrality is imperative and requires a drastic and rapid reduction in fossil fuel energy use [101]. On the other hand, it is worth noting that in other sectors of the global economy, the situation is not significantly better: In the "Industry" sector (responsible for 34% of total energy consumption), the RES share is 16.8%; in the "Buildings" sector (responsible for 34% of total energy consumption), the RES share is 15.9%; in the "Transport" sector (responsible for 34% of total energy consumption), the RES share is only 3.9% [39].

Figure 4 shows the changes in the absolute level of consumption of direct inputs (energy carriers) that took place in world agriculture over the period 1990–2021. In general, one can observe a marked increase in electricity consumption with a slight downward trend in the consumption of petroleum products and little change in other energy carriers. It is worth noting that during this period, for example, grain production increased from 1.95 bln to 3.07 bln tons (an increase of 57.4%) and population from 5.33 bln to 8.02) (an increase of 50.4 %) [102]. This suggests that the increase in demand for energy carriers was less than the increase in population and agricultural production (similar increases occurred in other product categories). Figure 5 represents the structure of energy carriers used in agriculture. The figures show that the most important agricultural energy carrier is petroleum products (diesel), accounting for about half of energy consumption. However, the situation in this regard varies somewhat between regions. The second most important energy carrier for agriculture is electricity. This structure is essential from the point of view of the possibilities of energy production by agriculture, as one can ask which types of renewable energy sources produced by agriculture can be a substitute for the energy carriers indicated in Figure 3.

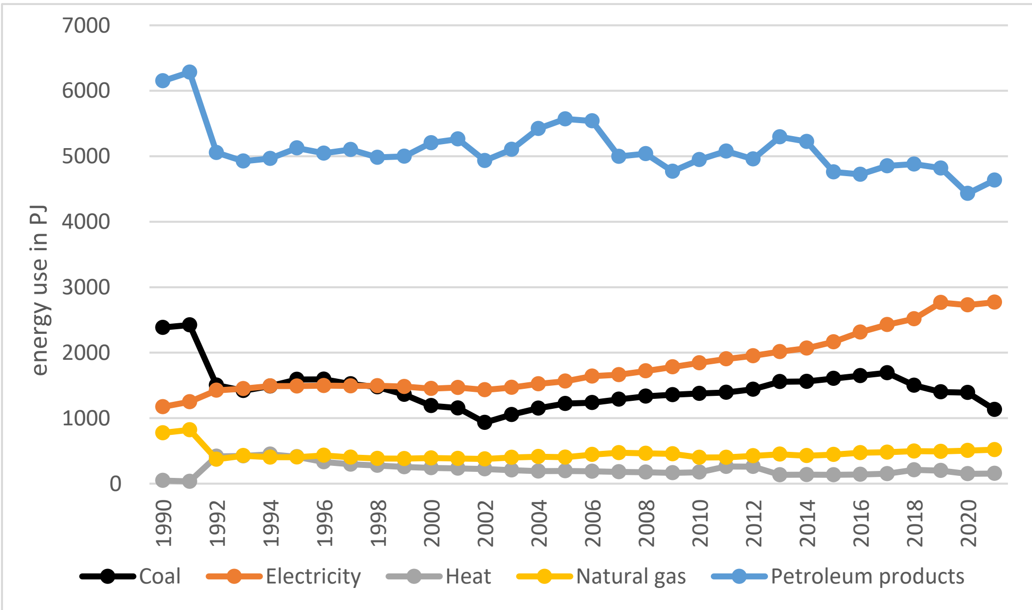

Figure 4. Global energy use in agriculture in the period 1990–2021 by energy carrier (direct inputs). Source: elaboration based on [103].

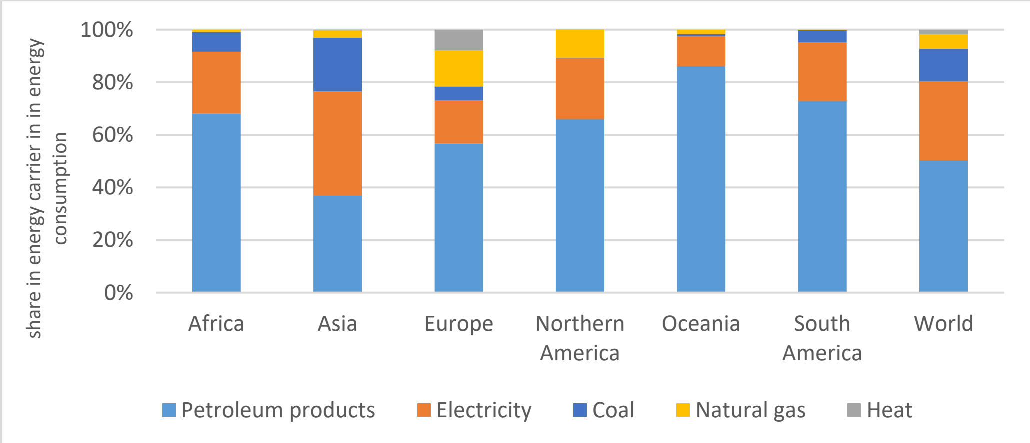

Figure 5. Structure of energy use in agriculture by in 2021 by energy carrier according to FAOSTAT (direct use). Source: elaboration based on [103].

A hallmark of modern agriculture, however, is its heavy reliance on industrial inputs, which means that the real significance of agriculture in energy consumption is greater than sector statistics indicate, and energy consumption reported, for example, in Eurostat statistics, could be underestimated [91]. Various estimates indicate that, in total, as much as 50% of total farm energy consumption may be due to indirect energy use included in fertilizers, feeds, and other inputs [91,93]. Estimates for the EU indicate that the share of energy in the costs of agricultural operations, on average, in the EU, for example, reaches 30% of the specific cost, 18% of intermediate consumption, and 12% of total input [104].

The energy intensity of the production processes of various inputs influences the level of indirect energy cost in agriculture. For example, according to the director-general of Fertilizers Europe, in 2022, the share of gas costs in the variable costs of fertilizer production reached as high as 90% [105]. This means that the agricultural sector strongly depends on the situation in the energy market, and energy price increases significantly affect agricultural production's profitability. The strong dependence of agriculture on energy prices means that in the case of low resilience of farms, the consequences of high energy prices can lead to a decline in production, which is an important issue not only from the point of view of farm economics but also from the point of view of global food security [106,107]. The problem is exacerbated by the fact that the global population is growing steadily. Despite the expected decline in the growth rate, this trend will continue until 2100, when, according to available forecasts, the earth will be inhabited by 10 billion people (against 8 billion currently). The "energy-agriculture nexus" thus addresses both energy supply and demand as well as food security and environmental (climate) challenges [86].

4.2. Energy Use and Type of Agricultural Activity

The literature lacks analyses based on detailed data indicating energy consumption in specific departments and production activities, including indirect inputs [28]. Analysis by Paris et al. [28] shows that for post-fallow crops in the EU, more than half of the energy consumption is related to using fertilizers and pesticides (50% and 3%, respectively). Diesel in such a mix accounts for 31% of energy, seeds for 6%, and other elements for about 8% (irrigation, storage, and drying are usually powered by electricity). Of course, there are differences between crop types and countries. The cited review also shows that in the group of arable crops, tillage is the most energy-intensive operation (from 43% to 61% in total direct inputs), followed by harvesting (from 14 to 31%), sowing (12–23%), fertilizer application (1–2%), pesticide application (1–7%), and other (up to 8%) [28]. The indicated activities refer mainly to diesel consumption. If the energy consumed indirectly is analyzed, then, as previously highlighted, fertilizer consumption dominates [28], whose production also generates high energy consumption [108,109].

In the case of livestock production, most of the energy-intensive processes take place in farm and livestock buildings (this applies to direct consumption). Electricity is used for lighting, feeding, and milking. Diesel is mainly used for manure management, and other fossil fuels (coal, gas) are used predominantly for heating [110]. The literature indicates that about 50 to 70% of direct energy consumption goes to diesel and the rest to propane, electricity, natural gas, and biofuels, although more thorough analyses are lacking [88,110,111]. Considering the indirect energy consumption of the livestock production subsector, the largest share of energy consumption in the agricultural sector is animal feeding [27,110,112]. Animal feeding is a process with particularly low energy efficiency due to metabolic and thermal processes, which means that most of the energy provided to animals in the form of feed is consumed at maintaining and renewing its cell structure rather than creating the final consumed "animal products" [110,112]. Moreover, the processes involved in the organization of animal feeding, such as transport, storage, and production of feed, also consume energy. Moreover, the raw material for feed production, i.e. mainly cereals, protein, and oilseed crops, also arises in processes that consume large amounts of energy [110]. These raw materials are sometimes transported over long distances from other parts of the world, which is also not indifferent to energy consumption [113]. An important factor shaping energy demand in the livestock production subsector is also the farm's location, which is particularly important in those activities that require heating or cooling of buildings—in colder regions, the demand is generally higher [114]. Considering the types of animal activities, some analyses show that the most energy-consuming subsector is milk production, followed by pig and broiler production [27,88].

Another data source for estimating the structure of agricultural energy consumption in the EU, for example, is FADN (Farm Accountancy Data Network). This database does not contain detailed information on energy inputs in physical units. Still, it contains highly detailed cost data, which indirectly makes it possible to assess the energy consumption structure. Table 2 shows the results of a review of FADN data [115] indicating the structure of energy costs incurred in EU agriculture. These data refer not to specific agricultural activities but to production types of commercial farms. Given that some of these types include specialized farms, it is possible to indirectly infer the activities that are most and least energy-intensive from the point of view of the entire sector (which is vital for energy and agricultural policy). The presented summary is based on the direct costs of energy carriers and the basic groups of indirect energy consumption, i.e., fertilizer and feed costs. The compilation shows that at the sector's scale, the largest share of total energy costs is accounted for by farms in the "field crops" type, which account for almost one-third of energy consumption (energy costs). Considering economic size (measured by standard output), it can be seen that the largest farms dominate this comparison. The second-largest farms in the total share of energy costs were those in the dairy type, accounting for 16.7% of energy costs, followed by farms in the "mixed" type. The largest farms were responsible for most of the energy costs in each farm type. Figure 6 presents the share of each production type in fertilizer costs, which, as the previous information shows, is the main item in indirect energy consumption. Considering the energy needed for fertilizer production, the nitrogen share reflects the energy consumption structure. The presented figure demonstrates that, as in the case of direct energy costs, farms in the "field crops" type are dominant, consuming approximately more than half of the fertilizers used in EU commodity agriculture. The next places in the ranking thus established are occupied by dairy and mixed farms, but their consumption share is several times smaller.

A little more detail on the energy intensity of various agricultural processes is provided by in-depth analyses conducted for EU agriculture as part of the AgEnRes project, which estimated the energy intensity of basic tillage operations [116]. The authors' estimate of the energy value of main direct energy inputs and fertilizers in the EU indicates that the most significant part of the energy consumed in agriculture is implemented in synthetic fertilizers (according to the cited estimate, it is about 35% of the total value included) (Figure 7). According to Paris [110,111], about 50% of expenditures on field production are spent on fertilizers. It is worth noting that greenhouse energy consumption accounts for less than 10% of total energy consumption in the EU. Although greenhouses have a significant energy demand, their share of the total sector is relatively small; hence, their contribution to total energy consumption is relatively low.

Table 2. Value of energy costs and share of specified farms' categories in total energy cost in the EU farming sector (direct and main indirect costs, i.e., fertilizers and feed). Source: own elaboration based on [115].

Categories of Farm Type (Specialization)	The Absolute Value of Energy Costs (mln EUR) in the Total Sector	Share of Specified Types in Total Energy Cost	Share of Specified Groups of Economic Size in Total Energy Costs of Specified Types	
			Farm Economic Size	Share (%)
Field crops	7576.5	32.6%	L	61.2
			M	24.3
			S	14.6
Horticulture	2341.4	10.1%	L	85.0
			M	12.3
			S	2.7
Wine	802.2	3.5%	L	64.3
			M	25.1
			S	10.5
Other permanent crops	1357.7	5.9%	L	34.4
			M	32.2
			S	33.4
Milk	3911.6	16.9%	L	83.8
			M	12.8
			S	3.4
Other grazing livestock	2519	10.9%	L	48.9
			M	37.4
			S	13.6
Granivores	2065.6	8.9%	L	97.1
			M	2.2
			S	0.7
Mixed	2634.4	11.4%	L	72.3
			M	15.5
			S	12.2

The results presented in Figure 8 indicate that plowing is the most energy-intensive operation, accounting for about 46% of total diesel consumption in crop production. In second place was harvesting (just over 30%), and in third place was crop maintenance (responsible for just over 15% of direct energy costs in crop production). The combined share of the three activity groups mentioned thus exceeded 90%. Also, in the case of crop production, the cited analyses indicate the dominance of several basic activities in the direct cost structure. “Feeding” and “manure removal” had the most significant and similar shares (both categories around 30%). Nearly half the share was for “heating and ventilation” (about 16.6%) (Figure 9). The share of other activities (watering, milking, air compression, cooling, water pumping, and others) was about 23%. It should be stressed, however, that the cited estimate refers to EU-wide averages, and there may be significant differences between regions [116].

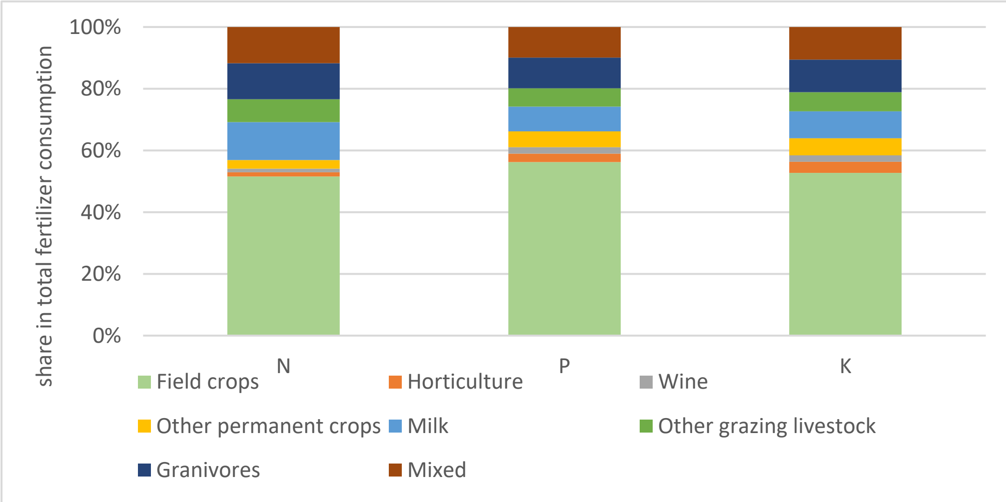

Figure 6. Share of specified categories of farms in total fertilizer consumption (proxy of energy indirect energy consumption). Source: own elaboration based on [115].

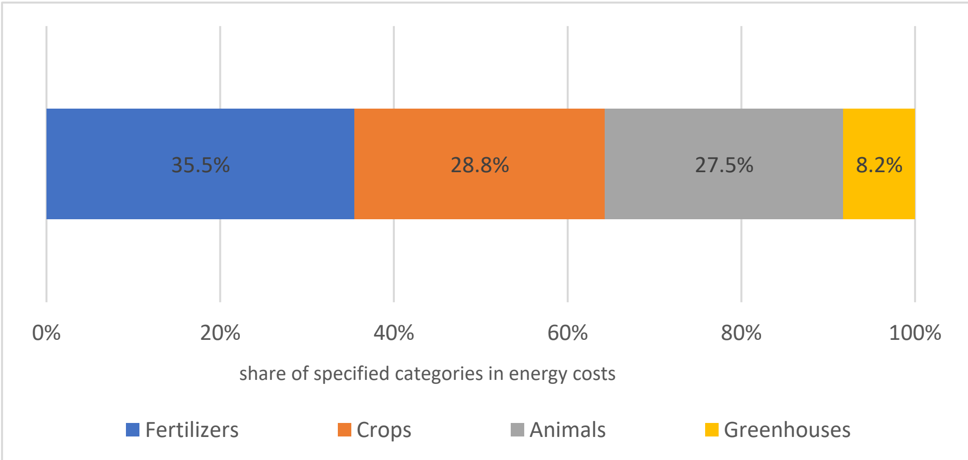

Figure 7. Structure of energy value of main direct energy inputs and fertilizers in EU. Source. Own elaboration based on [116].

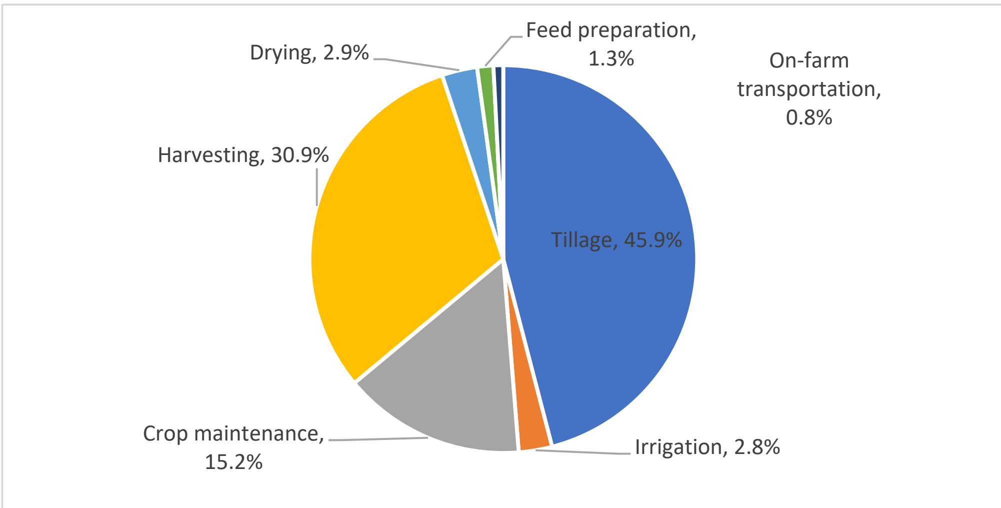

Figure 8. Structure of direct energy inputs in consumed diesel fuel for crop production. Source: own elaboration based on [116].

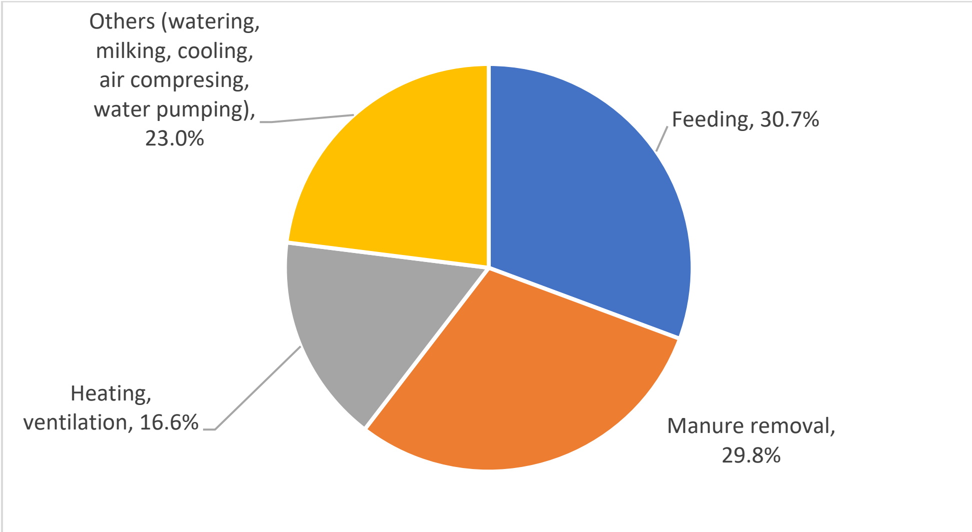

Figure 9. Direct energy consumption by main processes in total EU livestock production (without purchased concentrate feed and herd establishing in egg production). Source: own elaboration based on [116].

5. Improvement of Efficiency of Energy Use in Agriculture

5.1. General Information

As emphasized earlier, energy losses in agriculture refer to the inefficient use of energy carriers and the inefficient use of all means of production, which require energy inputs. The general formula for determining energy efficiency is the quotient of energy output and input [117–119]. The efficiency assessment should include direct energy sources (including manpower, animal energy, fuel energy, and electricity) and indirect energy sources (machine energy, seed energy, fertilizers, pesticides, and organic energy) [120]. Considering the energy contained in direct and indirect inputs and comparing this value with the energy contained in agricultural output leads to the conclusion that, on average, modern agriculture is inefficient. As indicated earlier, producing 1 calorie of energy requires as much as 10–15 calories in direct and indirect inputs [84,85].

Given that the primary factor of production in agriculture is land, it can be assumed that the area of agricultural land required to produce the various categories of agricultural commodities can serve as a proxy for indicating differences in the efficiency of energy use in different activities (virtually all production inputs in the crop subsector (diesel, fertilizers, pesticides), as well as a significant portion of inputs in animal production (feed), are closely related to land). Figure 10 shows the land required to produce 1000 kilocalories of specified products, which can indirectly measure energy efficiency. It is quite clear from this summary that the greatest demand for agricultural land (and consequently energy) is generated by the production of beef in beef herds. Combining milk and beef production in the dairy herds, the entire energy in the animal's life cycle is divided into beef meat and milk. Because of the assumption that agricultural area consumption can represent approximately energy consumption, it can be concluded that animal products are generally much less energy-efficient than crop products.

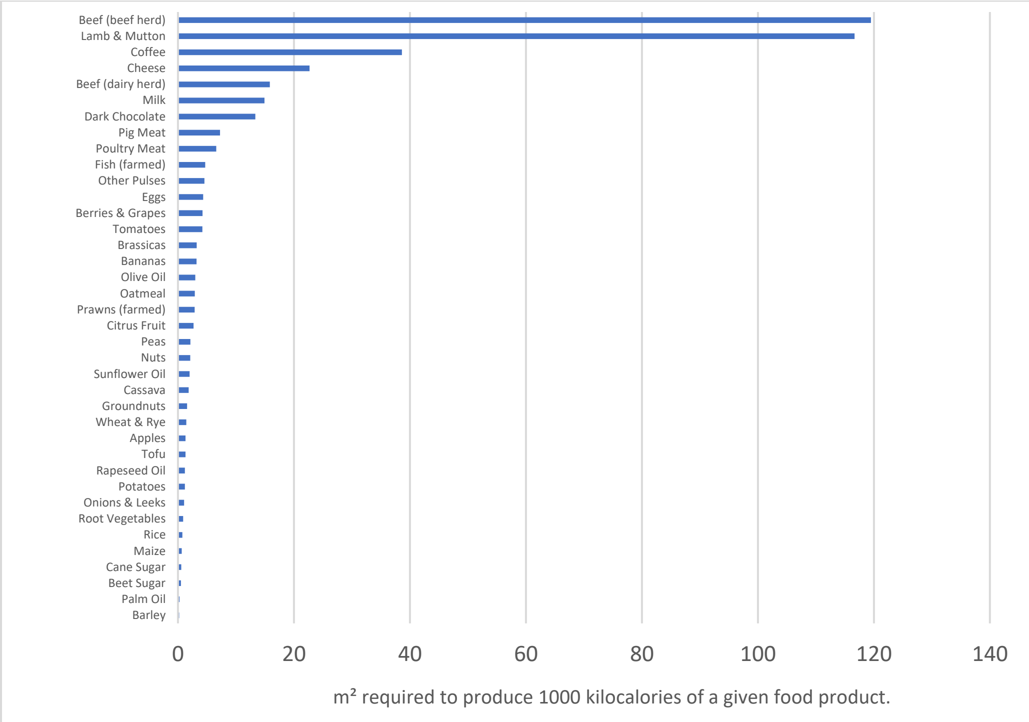

Figure 10. Land use in square meters (m^2) required to produce 1000 kilocalories of a given food product (a proxy of energy inputs). Source: [121] (additional calculations by Our World in Data (license CC BY)).

Given the problems associated with the degradation of natural resources as a consequence of inefficient use of energy in agriculture, among other things, as well as the need for energy transformation and a growing population, it seems obvious that we have to produce better food with less and clean energy [122,123], which means that energy use must be more efficient than before. Admittedly, OECD analyses [88] indicate that agricultural energy efficiency, measured as the ratio of agricultural GDP per unit of direct energy use, has improved since the early 1990s. Still, this process has occurred with rather large fluctuations and only in some countries. From a historical perspective, however, it can be said that the efficiency of using embodied energy in agriculture during the historical evolution of agricultural technology tended to increase as the efficiency of input generation processes such as electricity power generation, ammonia fertilizers manufacturing, or iron smelting improved [2]. Another way to improve efficiency processes is to reduce the demand for materials and energy due to the more efficient use of energy by machinery and equipment and the use of more efficient fertilizers or pesticides. Aguilera et al. [2] indicate that efficiency improvements in this area have recently declined because the production of many materials has approached thermodynamic limits. However, it is worth remembering that the energy efficiency of farms can be determined by a number of factors, which can be grouped into such categories as human factors, technological factors, organizational factors, and natural factors, among others [118].

5.2. Improvement Methods in Crop Production

When looking for ways to reduce energy consumption in the agri-food sector, it should be noted that the set of possible measures can be considered on several levels. Two basic ones include the farm level and downstream food supply chain [88]. The latter element is beyond the scope of the present considerations, so possible actions in this area will not be analyzed in the following section. However, it is worth noting that this area is diverse, consisting of the food processing industry, food retailing sector, restaurants, and catering [88]. The OECD indicates that activities to improve energy efficiency in this area generally fall under “management and monitoring”, although the set of specific activities to improve energy efficiency can vary widely (transport, chilled storage, lighting, cooking, ventilation, air conditioning, etc.). More complicated from the point of view of agricultural (farm) efficiency is the energy efficiency of input suppliers (e.g., fertilizers). The inputs supplied to farms make up indirect energy use; hence, the efficiency of their production processes translates into agricultural energy efficiency. In general, according to the OECD [88], energy efficiency improvements in the agricultural sector can be sought through “on-farm energy savings”, by “technical progress”, and through “substitution of inputs”.

It can be assumed that the greatest potential for efficiency improvement lies in the area of the most significant activities from the energy consumption standpoint. However, it should be noted that due to the diversity of production methods and types of agriculture, the importance of individual improvements will vary. Anyway, the most commonly identified ways to improve energy efficiency in the area of crop production include [116]:

- Application of precision farming techniques: This production system makes it possible to precisely dose fertilizers and pesticides, thereby optimizing the number of passes, which helps reduce fuel consumption (direct energy use) and also reduce the consumption of the inputs (fertilizers, pesticides), the production of which also requires significant energy inputs. Precision farming solutions are available mainly to large farms with a production scale that justifies the investment in costly technologies [101].
- Simplified tillage: Using minimum or strip-till techniques reduces the required trips and saves fuel. Tillage operations in the field are one of the most important components in diesel consumption, so keeping them to a minimum directly reduces energy consumption [124,125].
- Thoughtful management of tractors and machinery: This group of ways to improve efficiency is mainly related to diesel efficiency, and the range of possible measures varies greatly (aggregating machinery, combining conservation and fertilization, using tractors with horsepower adapted to needs) [126,127].
- Use of correct agricultural practices: for example, implementation of crop rotation and cover crops, which improve soil health and naturally reduce the need for mechanical soil tillage [101,128].
- Organizational improvements related to, for example, changing the production system (e.g., integrated, organic), which, by reducing the intensity of production, can reduce energy inputs, although the energy efficiency assessment is not always clear [88,92,129–131].
- Improving the skills of machine and tractor operators, especially in terms of economy and speed selection, keeping the machinery fleet in good shape [132].
- In the case of irrigation use, introducing energy-efficient pumps over better water management (ex., drip irrigation) generally improves water management and correlates with energy efficiency in crop production [133,134].
- Rationalization of transport processes (choosing optimal travel routes, transporting different things together to reduce the number of trips, etc.).

It is also worth mentioning that a possible option for improving efficiency in the area of crop production is the use of electric tractors. Full-electric tractors have a number of benefits. They are simple to integrate and to control; moreover, they allow for additional features, such tractive performance optimization in multi-motor configurations, and the production of zero local emissions. However, due to the low power density of the batteries (350 Wh/liter vs. 11 kWh/liter of diesel), it is expected that full-electric tractors at a small power size below 50 kW will soon arrive on the market [135]. For bigger tractors, which may require high power and endurance, various kinds of hybrid solutions are more feasible because of the higher energy density of diesel fuel and the possibility of fast refueling. It must be noted that the electrification of tractors provides some advantages, as previously used mechanical or hydraulic power supplies for tractor or machine components can be replaced by electric motors, which are easier to integrate and provide better control of operating parameters. The use of hydrogen-powered tractors could be pursued if more circular models for hydrogen production involving the farm itself were investigated [135].

5.3. Improvement Methods in Livestock Activities

The potential for improving energy efficiency in the case of livestock production seems to have been much less recognized so far. Given the importance of the livestock sector in energy consumption and how this translates into environmental impacts, measures to improve efficiency in this area are also essential. Given the livestock sector's significance in energy consumption and how this translates into environmental impacts, measures to improve efficiency in this area are also a no-brainer. As indicated earlier, animal feeding is one of the most energy-inefficient processes [110], accounting for up to 77% of total energy embodied in livestock [92]. On-farm energy consumption mainly includes housing and manure management [110,111]. Electricity consumption is connected to lighting, feeding, and milking, while diesel consumption is connected to manure management. What should be emphasized, however, is that the issue of energy efficiency in livestock production is poorly researched. There is a general lack of literature documenting energy consumption and energy efficiency in livestock operations [110,111]. In terms of improving efficiency in this area, one can emphasize, first and foremost, the sourcing of feed from raw materials with higher energy efficiency [27,88,110,111], bearing in mind that there may be some problems here, e.g., replacing cereal grains (which have high energy requirements) with grasses may improve the energy efficiency of livestock production, but at the same time may increase methane emissions from the stomachs of ruminants (but growing grasses has a higher potential for CO_2 sequestration, so, in the end, such an action will usually be beneficial both for energy efficiency of nutrition and for the climate) [88]. In addition, attention should be paid to such measures as limiting feed losses during animal feeding, carefully balancing feed rations, using underfloor heating in poultry rearing, recovering heat from ventilation systems, recovering heat from milk cooling processes, and using sunlight to illuminate buildings through transparent roofs. One possible way to improve efficiency is to rely on locally produced feed, which requires less energy [111]. The extent of possible energy savings and consequent efficiency improvements also depends on the type of operation, the correctness of production practices used, the location and condition of livestock buildings, and so on [136].

There are also specific solutions tied to different sections of animal production [116]. In the case of dairy cattle, one can consider increasing the role of pasture in the feeding system [137], the implementing of innovative grazing systems [138], changing the structure of crops used to make silage [139], abandoning feeds that require drying, optimizing the use of concentrates and other feeds, as well as the use of properly balanced feeds [140,141], optimizing transport, improving the efficiency of washing milking systems, using the heat from cooled milk [142,143], and optimization of cow-milking processes [144]. In the case of pig production, in addition to the universal recommendations related to improving the use of feed, the following can also be mentioned: improving water management [145], optimizing the use of heat energy in piglet buildings [146], reducing energy consumption for lighting, maintaining optimal microclimate, etc. [147]. In the case of poultry production, proper feed management, prevention of heat loss through proper insulation, and efficient and economical ventilation systems are also crucial [148,149], or the use of more efficient floor heating [150] and heat recovery [149], reducing uncontrolled heat loss and controlling the microclimate [151]. It is worth noting that activities that contribute to the reduction of energy consumption (post-consumption) also support the reduction of greenhouse gases. However, it is important to remember that livestock production is strongly linked to methane emissions from enteric fermentation in ruminant organisms and from manure management. Taking measures to reduce methane emissions, such as managing manure in a biogas plant or optimizing nutrition, also improves energy efficiency on farms, and what is more, the biogas extracted reduces GHG emissions in two ways, i.e., it prevents methane emissions and also replaces fossil fuels [152–154].

Thus, it can be concluded that some of the measures to reduce energy consumption in livestock production can also contribute to reducing greenhouse gas emissions. This includes both CO_2 emitted from the burning of fossil fuels as well as a reduction in methane emissions due to better manure management or improved animal nutrition. [155,156].

5.4. Modern Technical Solution

Recently, more and more effects of improving the efficiency of energy have also been seen in the concept of smart farming and the use of artificial intelligence [67,136]. Smart farming, in relation to the problem of energy efficiency, enables better management of energy sourced from own sources (e.g., photovoltaics integrated with a wind turbine) through energy storage and automation that monitors and manages the farm's energy system. A case study presented by Pereira et al. [126] showed that with such solutions, it was possible to reduce energy draw from the grid—for example, farms in central Portugal, by as much as 83%.

It is also worth highlighting the potential for reduction due to technological advances and increasing availability of precision equipment. It is particularly worth highlighting physical methods and technologies in agriculture supporting energy efficiency. Among these methods are [157] physical technologies and materials for the passive management of the solar spectrum in greenhouses, laser remote sensing in agriculture, laser-induced breakdown spectrometry for rapid chemical analysis in agriculture, application of polarimetry in biological and agricultural diagnostics, plasma technologies, or optical methods for detecting phytopathogens. The referenced solutions support precision production, which facilitates management and reduces production inputs (or increases outputs), improving the energy efficiency of agriculture. Precision agriculture makes it possible to reduce inputs through precise dosage, thus reducing indirect costs. It can also be important, for example, to reduce fuel consumption by optimizing trips. The use of smart farming helps monitor and optimize various processes [158,159].

When considering opportunities to improve energy efficiency, it is also important to pay attention to possible infrastructural adjustments (including, in particular, in greenhouses) such as insulating buildings, upgrading ventilation systems, heat recovery from other processes, reducing uncontrolled heat loss, etc. [116,160,161].

5.5. Reduction of Food Waste and Losses

Undoubtedly, one important area of action to improve efficiency is also to reduce waste. It is estimated that food loss and waste account for about 8–10% of global annual emissions of GHG [162]. Recent estimates indicate that a total of 40% of food is never eaten [31]. It is also estimated that more than 15% of global losses are related to the farming stage [31]. In general, it can be observed that the higher the degree of economic development, the greater the per capita food losses, the greater the share of food losses at the consumption stage, and the smaller the share at the farming and processing level [29]. The lower level of agricultural development in developing countries makes losses on this link particularly high. Major loss sources in the Sub-Saharan Africa region, for example, include harvesting and field drying (4–8%), transport to homestead (2–4%), drying (1–2%), threshing/shelling (1–3%), winnowing (1–3%), farm storage (2–5%), transport to market (1–2%), and market storage (2–4%) [163]. Ultimately, however, recent estimates indicate that 58% of global farm-stage food waste is generated in middle- and high-income regions [31]. Reducing food waste in agriculture is crucial to enhancing energy efficiency. Possible measures include improving storage and harvesting techniques, although this may require investment in more precise machinery and equipment. Adequate harvesting schedules are also important, as delays in this regard can lead to deterioration in product quality and consequently increase wastage. Likewise, proper crop protection and fertilization techniques can help maintain shelf life (both over- and under-application of fertilizers/pesticides can be used to increase product wastage during storage). Another important measure is transportation under proper conditions (temperature, humidity), as many agricultural products are sensitive to transportation parameters. The same applies to all warehouses and places where agricultural products are stored [164]. From the point of view of farm management, it is also important to prepare plans for production (including the selection of more resistant varieties) and sales, since overproduction can involve the difficulty of disposing of products in a timely manner, which can also lead to their wastage. Of course, all precision farming techniques are useful, not only in helping reduce inputs but also in conducting production in a more controllable way, reducing the impact of adverse environmental factors and resulting in higher yields and quality of crops [165]. From the point of view of the agri-food system as a whole, it is also important to use food products that are no longer fit for consumption for energy production or to process them in other ways (such as composting) to prevent unproductive energy loss [166].

6. Agriculture as an Energy Producer

Combined with energy conservation practices, farmers can generate their own energy to become even more self-sufficient, reducing external inputs [39]. Renewable energy not only helps farmers save money but also helps fight against global warming [104]. Agriculture can generate renewable energy beforehand based on biomass, solar energy used to produce electricity in PV panels, and wind energy [5,38,39,167]. The potential contribution of agriculture to renewable energy production is schematically depicted in Figure 11. So far, however, using renewable energy from agriculture in the overall economy remains relatively small, although no accurate statistics present agriculture's role in providing energy. However, given that the total global renewable energy in total final energy consumption is estimated at about 15.4% [39], there is no doubt that the contribution of agriculture itself to this value must be significantly lower. Report REN21 [39] indicates that outside the EU, only four countries (Bangladesh, India, the Republic of Korea, and Malta) have renewable energy targets for the agricultural sector, despite the myriad benefits of integrating renewables with farming practices.

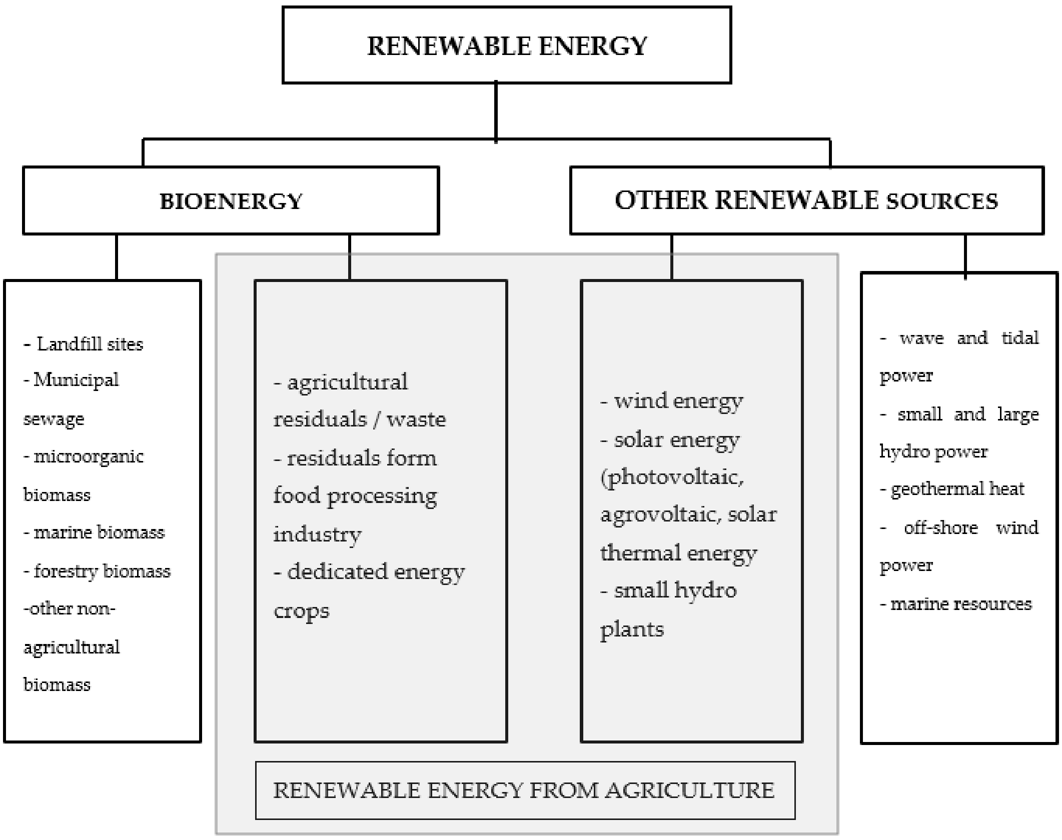

Figure 11. Categories of renewable energy from agriculture. Source: elaboration based on [168–171].

6.1. Biomass

Biomass is the primary way in which agriculture can obtain energy, and historically, it is the oldest form of energy extraction by mankind [2,3]. The literature indicates that the use of agricultural biomass to produce the generation of various energy carriers is probably the most important option when it comes to increasing agriculture's involvement in energy generation, both for consumption in the agricultural sector itself, as well as for the needs of other sectors of the economy [104,116]. Energy extracted from biomass (organic materials) and converted into electricity, transportation fuels, and heat is referred to as bioenergy [172]. Globally, biomass energy is estimated to contribute 10–14% of total energy demand [95]. Statistics from the World Bioenergy Association [173] indicate that bioenergy in the World energy mix accounts for about 9%. The value of the entire global bioenergy market is estimated at USD 296.7 billion, with CAGRs (Compound Annual Growth Rates) of 9.4% over the next few years [174].

Overall, biomass dominates the structure of RES in the European Union, accounting on average for almost 60% of the share in RES, and in some countries, it even exceeds 90%. Significantly, however, the main contribution to this result comes from woody (forestry) biomass (about 60% of biomass) [175,176]. The share of agricultural biomass, which includes agricultural by-products and agricultural crops, is estimated at about 27% of the total biomass (the rest is municipal, etc.). However, harvesting the currently dominant woody biomass will be justified if the process is sustainable and reinforces the need for carbon dioxide sequestration and meeting biodiversity goals [177]. Thus, the potential for woody biomass in the EU can be expected to decrease [175]. Given the growing challenges and requirements of the circular economy, agricultural biomass (second-generation substrates) appears to have much greater potential [178,179]. Agricultural biomass has many advantages as a source of renewable energy. Among the most important of these are the utilization of waste and residues from the farm sector, which allows the reduction of emissions from agriculture, diversity, decentralization of energy supply, and improvement of energy security at the regional level, and the possibility of producing different energy carriers and different products in agricultural biorefineries or the creation of new jobs in rural areas [116,175].

According to Bioenergy EU, only about 15% of the biomass used for energy production in the EU in 2020 came from agriculture (this figure consisted of agricultural residues and energy crops for heat, electricity, and biofuel production [180]). Bioenergy Europe stresses, however, that agricultural biomass is underutilized due to a lack of adequate policy measures, limited knowledge, or appropriate business models. Also, other studies indicate that agriculture's potential for renewable energy generation opportunities is underutilized, with farms making little use of their own energy [181]. The significant potential of agriculture in bioenergy production is because only a portion of agricultural products are consumed, and depending on the commodity, residues can account for 10 to as much as 90% [182].

A study by Majewski et al. [183] of the agricultural biogas potential from substrates derived from animal production in the case of Poland showed that it would be possible to cover the demand of the agricultural sector for electricity. Biomethane also has great potential; compared to biogas, this technology has an advantage, as it allows for better energy use. Some prognoses indicate that by 2050, agricultural biomethane will meet up to 30 to 40% of the EU's gas needs (so far, biogas and biomethane have been able to meet about 5% of the EU's gas needs) [184,185].

Some authors note [175] that only a part of the potential contained in biomass from livestock production can be used to generate energy due to the need to fertilize fields. Still, it is worth adding that the digestate, such as manure, obtained in biogas plants has good fertilizer properties [58,186].

Biomass can generally be assigned to one of three categories, i.e., plant, animal, and microorganic biomass [187]. Agricultural biomass can be used to generate energy in several ways [37]: as a directly combustible fuel, as gaseous fuel (biogas), as liquid fuel (ethanol, biodiesel), as fuel pellets, and for generating electric power. A variety of fuels (liquid, solid, and gaseous energy carriers) such as biochar, biodiesel, bioethanol, biobutanol, biogas, biomethane, and biohydrogen can be produced from biomass [188,189]. Depending on the type of biomass from which bioenergy is extracted, four generations of biofuels can be distinguished, i.e., first-, second-, third-, and fourth-generation biofuels [190,191]. First-generation biofuels are derived from agricultural crops: grain crops, beets, corn, etc., produce bioethanol, while oil crops (oil palm, soybean, coconut, rapeseed, and sunflower) produce biodiesel [188,190]. In general, first-generation biofuels are produced from edible food sources, which generates competition for agricultural land, and the legitimacy of this solution is fraught with great doubt [192]. Of particular importance is that future biofuel generations will not use edible food resources. Second-generation biofuels primarily include energy carriers derived from the decomposition of lignocellulosic biomass and other waste streams, including various organic wastes, particularly agricultural residues, such as those from the food industry, such as wheat bran, animal fats, or waste from cooking and frying oil [190]. It should be emphasized that according to the EU RED II Directive, biomass for sustainable energy production should not create competition for agricultural land devoted to food and feed production [184]. Between 2010 and 2015, nearly half of the newly installed biomethane plants were based on energy crops [184]. Third-generation biofuels are based on the use of marine biomass, particularly macroalgae and seagrasses, while fourth-generation biofuels are based on the use of genetic engineering to enhance the desirable traits of organisms used in biofuel production [190]. Agriculture's bioenergy production potential is based mainly on second-generation biofuels, i.e., the use of waste agricultural raw materials to produce, for example, agricultural biogas. However, they can also be used to produce other energy products and materials in so-called agricultural biorefineries [58].

A particularly useful technology for using agricultural biomass is the production of biogas, including upgrading to biomethane [185,193]. Agricultural biogas produced from organic fertilizers (manure, slurry) not only enables the replacement of a certain amount of fossil fuel energy with renewable energy but also reduces methane emissions that occur during traditional storage of manure and slurry [22,152,194]. It is assumed that a 1 kg reduction in CO_2 emissions from replacing fossil fuels with agricultural biogas produced from organic fertilizer translates into an additional 1 kg reduction in CO_2 emissions due to a change in fertilizer management [153].

The efficiency of energy generation in biogas plants varies depending on how the biogas is used. For example, for combined heat and power (CHP) generation, the typical electrical and thermal efficiencies are 40 and 50%, respectively (so the key to efficiency is whether the thermal energy generated in the combustion engine can be managed). In many cases, it is impossible to efficiently harness thermal energy, which means that part of the primary energy is dissipated [184]. In contrast, for thermal energy generation by burning biogas in a gas boiler, the value is 82.5% [195]. If biogas is upgraded to biomethane, it can be used in various ways, just like natural gas, and the efficiency of this process will depend on the processing technology [185]. Biomethane has a large potential for reducing GHG emissions: It can be assumed that replacing fossil fuels with biomethane reduces emissions by 80%, and reductions of 200% are possible in some solutions (in addition to the emissions avoided from replacing fossil fuel, a similar amount of GHG is effectively removed from the atmosphere and stored in the land or can be avoided in adjacent systems) [196]. As a result, lifecycle emission is expected to be below zero [195]. Another benefit of biogas and biomethane plants is that compared to other renewable energy sources, biogas plants are characterized by greater production stability [197]. Moreover, biogas production supports the distributed energy model, which is small-scale, decentralized energy generation focused on local use, which fits in with prosumer energy [198–200]. From this point of view, micro biogas plants are an interesting solution; they can be established even on relatively small farms and used by farmers for their own needs (energy self-sufficiency). Such facilities are usually powered only by biomass from a given farm (group of farms), eliminating the need to transport substrates and making them independent of substrate price fluctuations [153,201].

6.2. Agrovoltaics (Photovoltaics)

Generally, it can be assumed that growing crops as a food source for humans and animals involves converting solar energy into biomass [85]. However, the sun's energy can also be used to generate electricity in PV panels, and farmers can do this as well—partly for their consumption, while the unused surplus can be directed to other consumers [167]. In recent years, there has been a marked increase in interest in photovoltaics in the agricultural sector, with the development of the concept of agrovoltaics [202]. Agrovoltaics is a solution that allows a given area of farmland to simultaneously grow crops and produce energy from PV panels [203–206], in addition to the apparent benefit associated with the dual use of agricultural land (growing crops and generating energy agrovoltaics offer farmers other benefits related to reducing water consumption or protecting the plants under the panels from extreme weather conditions). On the other hand, however, it should be borne in mind that PV panels restrict the plants' access to light, which limits their productivity. Analyses by Dupraz [207], however, show that through the two-way use of land, its productivity increases in the 35–75% range, depending on the densities of PV panels. In contrast, a 30% increase in economic productivity is indicated by the analyses of Dinesh and Pearce [208]. An important prerequisite for successfully using agrovoltaics is the selection for cultivating plants that tolerate shade well [208]. Therefore, agrovoltaics associated with livestock grazing or pollinator habitat seem more financially competitive than agrovoltaics based on crop production [39]. A particular form of agrovoltaics is the photovoltaic greenhouse (PVG) [209,210].

Mention should be made of hybrid solutions, such as integrating wind energy with biomass-based or photovoltaics (agrovoltaics) with other sources [67,211]. It is also worth mentioning that solar PV can be used as a thermal energy source (both for heating and cooling)—a particularly useful solution in those agricultural activities where hot water is needed (greenhouses, animal buildings) [39,207]. It is also possible to combine solar PV with thermal technologies.

6.3. Wind Energy

Agriculture also has untapped wind energy opportunities [55]. Unlike solar power, wind power is available 24 hours a day, but there is also a strong dependence on atmospheric conditions. Hence, not all regions are predisposed to this energy generation method [38]. The turbine's work there is that the rotating propellers power an electric generator, which turns the wind's kinetic energy into mechanical or electrical energy [212]. Historically, many centuries ago, wind energy was used to power "wind pumps" to irrigate or drain fields [213]. Today, electric turbines dominate. Wind turbines can vary significantly in power output (from a few kW to even several MW (megawatts)); hence, they can provide energy to farmers of different scales of operation. Small-scale wind turbines became popular in Texas, for example, as early as the 1930s and 1940s [214]. They have been used, among other things, to pump water for cattle in areas without access to the power grid. Turbines installed on farms can vary significantly, and one important advantage is the small area required to install the equipment [38].

The energy extracted from wind turbines can be used for various purposes, ranging from powering irrigation systems to producing electricity. Farmers can participate in the production of wind energy in three ways, i.e., taking up a partnership with a "wind developer" and allowing the installation of a turbine on their land; erecting their turbine partly for their own use and partly for energy sale; or becoming a "wind developer" themselves and focus solely on the energy business [214].

7. Major Challenges—An Attempt at Synthesis

The above review of issues and problems shows that the agricultural sector should be much more involved in renewable energy production than it has been so far, which is important both from the point of view of farm efficiency and meeting global climate policy goals [215]. Achieving these goals requires the involvement of all countries of the world and all sectors of society, according to their capabilities [32]. Agriculture, as a producer of biomass, has significant opportunities to join in the generation of useful forms of energy: It is estimated that the share of "modern bioenergy" (i.e., without traditional biomass combustion) in final energy consumption will increase globally from about 3% in 2018 to 18% in 2050 [55]. Agriculture has an essential contribution to make to this process. Moreover, agriculture also has the opportunity to participate in generating energy from photovoltaics (agrovoltaics) as well as wind power. So far, however, the transition has been slow, and agriculture itself is globally based mainly on fossil energy [39]. As FAO and IRENA point out [55] (p.65), transformation of the agri-food sector to renewable energy requires "concerted efforts between government, the private sector, financing institutions, academia, and international and non-governmental organizations". Economic factors related to the support of prosumer solutions play a particularly important role. Different regions of the world have different experiences and opportunities in this regard, but it should be emphasized that the key challenge is to shift energy subsidies from fossil fuels to renewable energy [216]. In 2020, around USD 5.9 trillion was spent globally to finance the fossil fuel industry (through explicit subsidies, tax breaks, and health and environmental damages that were not priced into the cost of fossil fuels) [216]. It can be estimated that even a portion of this amount reallocated to support bioenergy generation would significantly accelerate the increase in bioenergy use. Over the past several years, there has been a significant decline in capital costs and LCOE (levelized cost of electricity) for sources such as solar PV and wind energy, and a slight decline for bioenergy (Figures 12 and 13). This means that financially, bioenergy production is less competitive, but has an advantage because of the stability of production, which is important both for the consumers themselves and for the stability of the power grid.

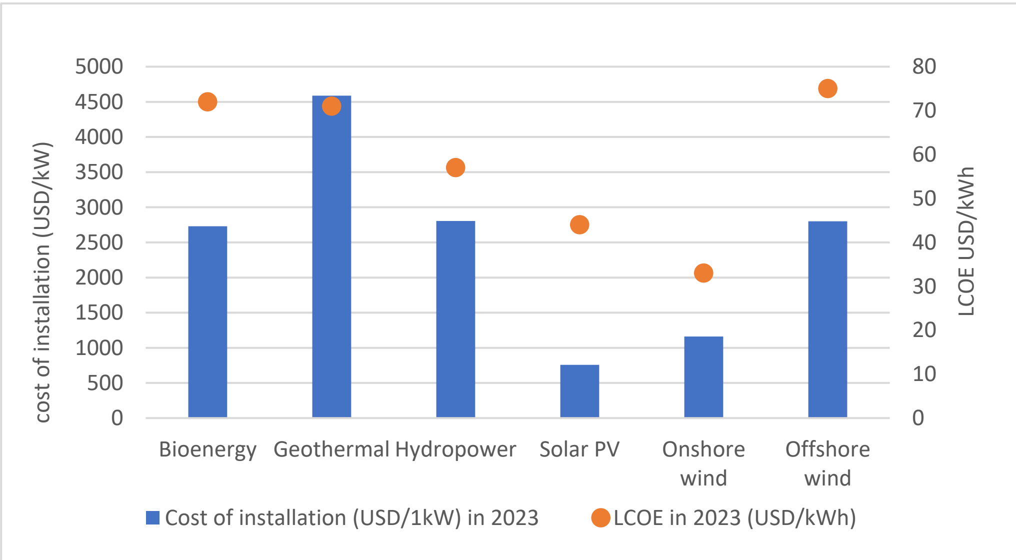

Figure 12. Cost level of installation of various RES (in USD/1kW) and LCOE (in USD/kWh). Source: elaboration based on [74].

It is worth noting that various forms of support for prosumers and renewable energy production are known, such as tax incentives, feed-in tariffs, different categories of certificates, and the changing costs of renewable energy generation, which require adjustment of support policies to current economic conditions [217,218]. However, volatility in support policies causes investment instability, making investment decisions difficult. Creating policies that facilitate investment and reduce risk, as well as removing barriers to access to materials and technology, is also an important challenge [216]. Table 3 summarizes the key factors that, based on the literature review conducted, can be considered enablers and barriers in the process of increasing agricultural involvement in renewable energy production, particularly based on biomass.

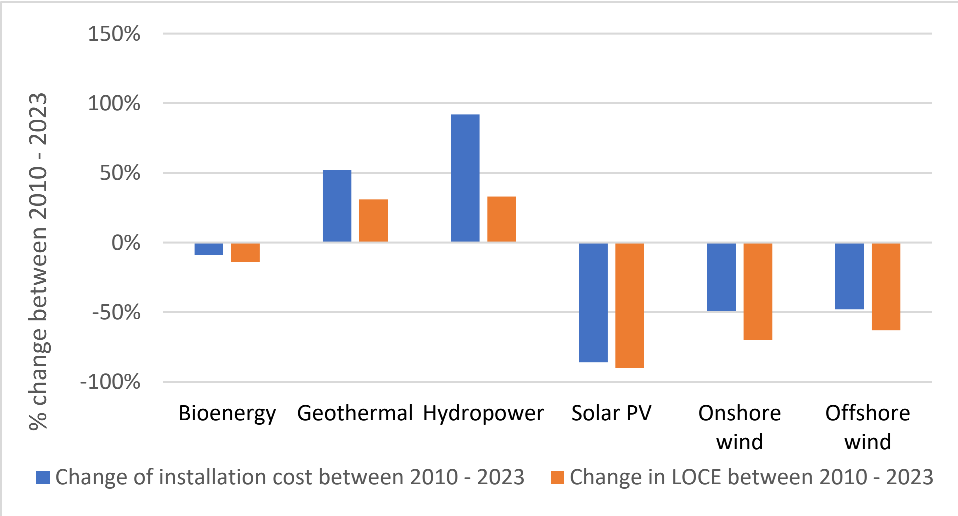

Figure 13. Percentage change in installation costs and LCOE of specified RES in the period 2010–2023. Source: elaboration based on [74].

Table 3. Enabling factors and barriers to the development of renewable prosumer energy in agriculture (based on bioenergy). Source: own elaboration.

Factors Stimulating the Development of Renewable Energy in the Agricultural Sector	Barriers and Threats to the Development of Renewable Energy in the Agricultural Sector
Global climate policy assumptions aimed at providing "clean energy at an affordable price"	Climate policy goals in the field of energy can be achieved using other solutions
Stronger integration of agriculture into the production of renewable energy based on biomass (especially biogas) enables the reduction of greenhouse gases by replacing fossil fuels as well as by changing the management of manure or other biomass sources.	Climate policy in agriculture will increasingly exert pressure to reduce animal production and reduce food losses, limiting the availability of second-generation substrates. Effectively using agricultural biomass in prosumer energy is also difficult due to the dispersed agrarian structure (it is difficult to use scale effects).
New technologies for obtaining energy from third and second-generation biomass are being developed.	The use of third- and fourth-generation biomass tends to extend beyond the agricultural sector and the prosumer energy model.
Fossil fuels used in agriculture are still heavily subsidized in many regions of the world, which provides an opportunity to shift support towards renewable energy, including in agriculture.	The fossil fuel-based energy industry is not necessarily interested in change; energy companies are often state-owned and constitute an important source of government re-venue.
The development of renewable energy leads to a gradual reduction in investment costs and LCOE	Technologies based on bioenergy (biomass), which is particularly important for agriculture, remain among the most expensive. In this case, the cost decline over the last years has been much smaller compared to PV and wind energy.
The production of renewable energy in agriculture is an opportunity to reduce energy costs on farms and increase their energy independence.	Profitable investments in modern solutions usually require public funding, which depends on policymakers.
Investments in renewable energy sources in agriculture support the dissemination of the distributed energy model.	Poor infrastructure, low quality of the network, and lack of connections make the distribution of energy (biogas) difficult or impossible.

In view of the fact that renewable energy sources are less stable and production is less controllable than in the case of fossil fuels, a major challenge for agriculture is also the issue of harvesting energy. This is particularly true for solar and wind energy, where the daily profile of radiation availability often differs significantly from the profile of electricity demand. In the case of biomass converted to biogas, the problem is less severe, as biogas (as well as biomethane) can be stored relatively easily (at least in the short term). In the case of electricity, farmers, like other prosumers, can send surplus energy to the grid [153]. However, the realistic possibilities depend on the capacity of the power grid and the economic viability of the billing system with the energy buyer (net-metering or net-billing). The experience of many countries (Germany, Australia, USA) shows that the transfer of energy to the power grid with net-metering billing generates, among other problems, increased electricity prices for non-prosumers, voltage fluctuations, overvoltage and undervoltage, instability, and reverse power flow [219]. In view of the existing problems, it makes sense to promote on-farm energy storage to manage energy consumption and production better. Energy storage is a possible solution, but at the current stage of development, it involves large capital expenditures. Another possible solution is micro-pumped hydroenergy storage using agricultural reservoirs, which, however, are useful only in certain locations [220]. It can be argued that while in recent years, there has been great progress in the spread of renewable energy technologies, which has translated into a decrease in their costs (at least some), the weak point remains precisely the accumulation of created energy so that it can be used when it is needed.

8. Conclusions

Global climate policy clearly puts pressure on the search for ways to operationalize ambitious energy transition targets. This process requires changes in many sectors of the economy, including agriculture. Despite agriculture's relatively small share of direct energy consumption, the agri-food sector negatively impacts the environment in many different ways, which can be seen as a result of the inefficient use of various forms of energy. At the same time, agriculture has significant opportunities to contribute to the production of energy, both for its own use and for the needs of other sectors. Particularly, high hopes can be pinned on using the emerging biomass sector in agriculture and agri-food. However, farmers can also be involved in producing solar PV (agrovoltaics) and wind energy. If global energy transition targets are to be realized, we should look at agriculture as a specific kind of "energy prosumer." Realizing the potential of agriculture to meet the goals of the energy transition toward a sustainable model requires reducing the consumption of direct and indirect energy, improving the efficiency of the use of energy consumed, and involving agriculture in the production of renewable energy for its own use and for other sectors of the economy.

First, enhancing energy efficiency in agricultural practices is essential. This can be achieved by minimizing energy-intensive operations such as tillage and optimizing the application of fertilizers in crop rotational systems and feed used in livestock production. Integrating technologies such as precision agriculture and smart farming can facilitate these enhancements. Second, scaling up renewable energy production within the agricultural sector is imperative. Viable strategies include the utilization of biomass, the integration of agrovoltaics (solar panels within agricultural landscapes), and the harnessing of wind energy. Notably, converting waste biomass to biogas holds great potential as a substitute for fossil fuels and as a measure to mitigate methane emissions from decomposing organic waste. Despite these possibilities, the agricultural sector remains predominantly reliant on fossil fuels, with relatively limited uptake of renewable energy technologies. Increased participation in energy production could position agriculture as a pivotal player in sustainable energy systems, paralleling the role of prosumers who actively engage in energy generation, consumption, and storage. However, progress is hindered by several challenges, including low adoption rates of renewable technologies, a fragmented landscape of research efforts, and a lack of comprehensive assessments regarding energy efficiency and the bioenergy potential within the agricultural sector.

The transition to prosumerism in agriculture presents several significant challenges for the future. These challenges thus define directions for future research. A primary concern is energy storage, as renewable sources such as solar and wind produce energy intermittently. To effectively balance production and consumption, it is essential to develop affordable and efficient storage solutions, including advanced battery technologies. Next-generation biofuels promise a cleaner energy landscape; however, their advancement is contingent upon overcoming technological and economic barriers that currently limit scalability. Furthermore, aligning agricultural energy production with the Sustainable Development Goals (SDGs) and climate policies is a multifaceted challenge. This alignment necessitates carefully balancing increasing energy demands with environmental conservation efforts, equitable resource distribution, and commitment to climate action objectives. An important challenge requiring scientific research is the search for ways to use modern technological solutions on a wider scale. Modern science knows of many methods for more efficient energy use that fit into the concept of precision agriculture and smart farming. Still, the degree of their dissemination remains insufficient due to organizational and economic barriers. Research on this issue appears crucial to the spread of technological advances in many regions of the world. Integrating these aspects while addressing socioeconomic constraints should help to transform agriculture into a sustainable energy prosumer sector.

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